SMART INDIA HACKATHON 2026

UNIGUARD AI

Passive Unidirectional Cyber Threat Intelligence Platform

SMART INDIA HACKATHON 2026

OFFICIAL PROBLEM STATEMENT DETAILS

Problem Statement ID: SIH26145

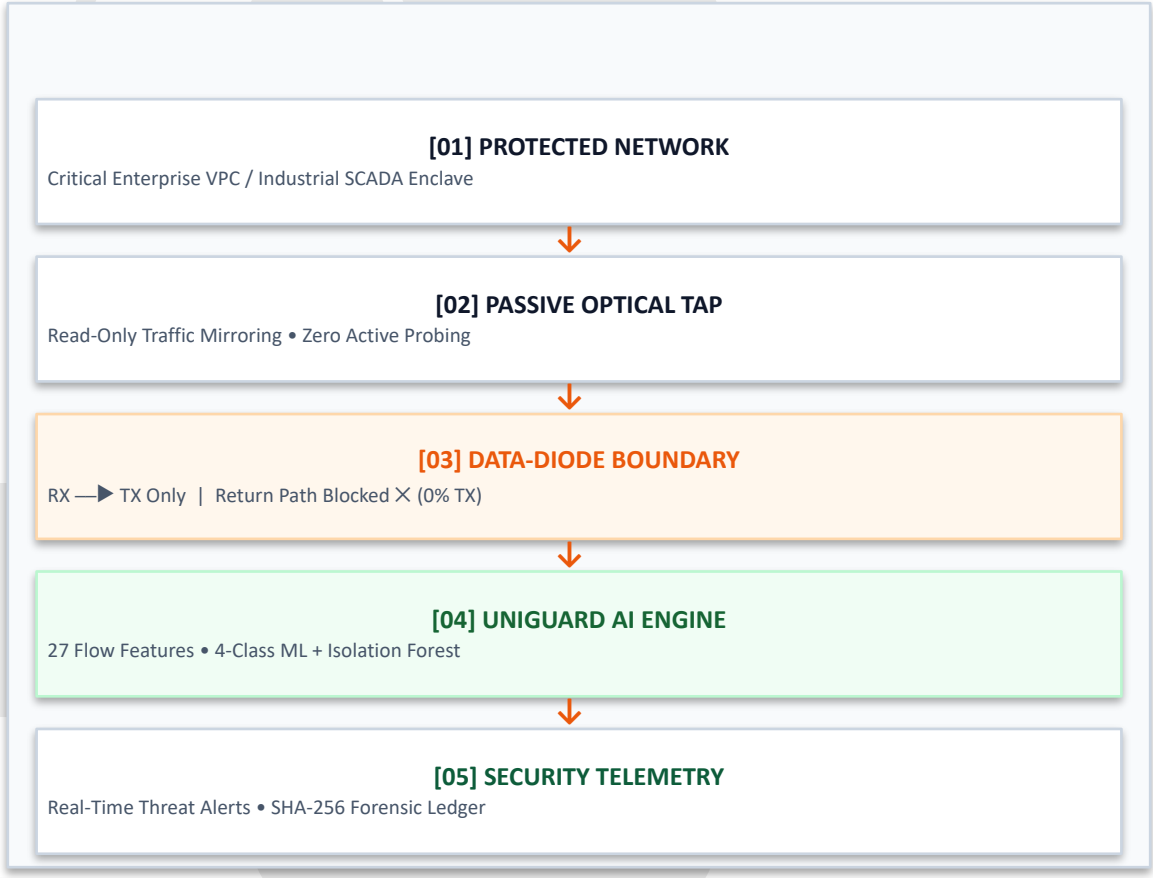
Problem Statement: **AI-Based Detection of Cyber Threats in Unidirectional IP Traffic**

Theme: Blockchain & Cybersecurity

PS Category: **Software**

Team Name: **TEAM LIQUID (Rathinam Technical Campus)**

Monitored Enclave: Critical Infrastructure / Defense OT / SCADA Enclaves



TEAM LIQUID

ED SOLUTION

ECTURAL INNOVATIONS & END-TO-END UNIDIRECTIONAL FLOWCHART

SMART INDIA HACKATHON 2026

CORE ARCHITECTURAL INNOVATIONS

01 | PASSIVE BY DESIGN — Zero Active Probing

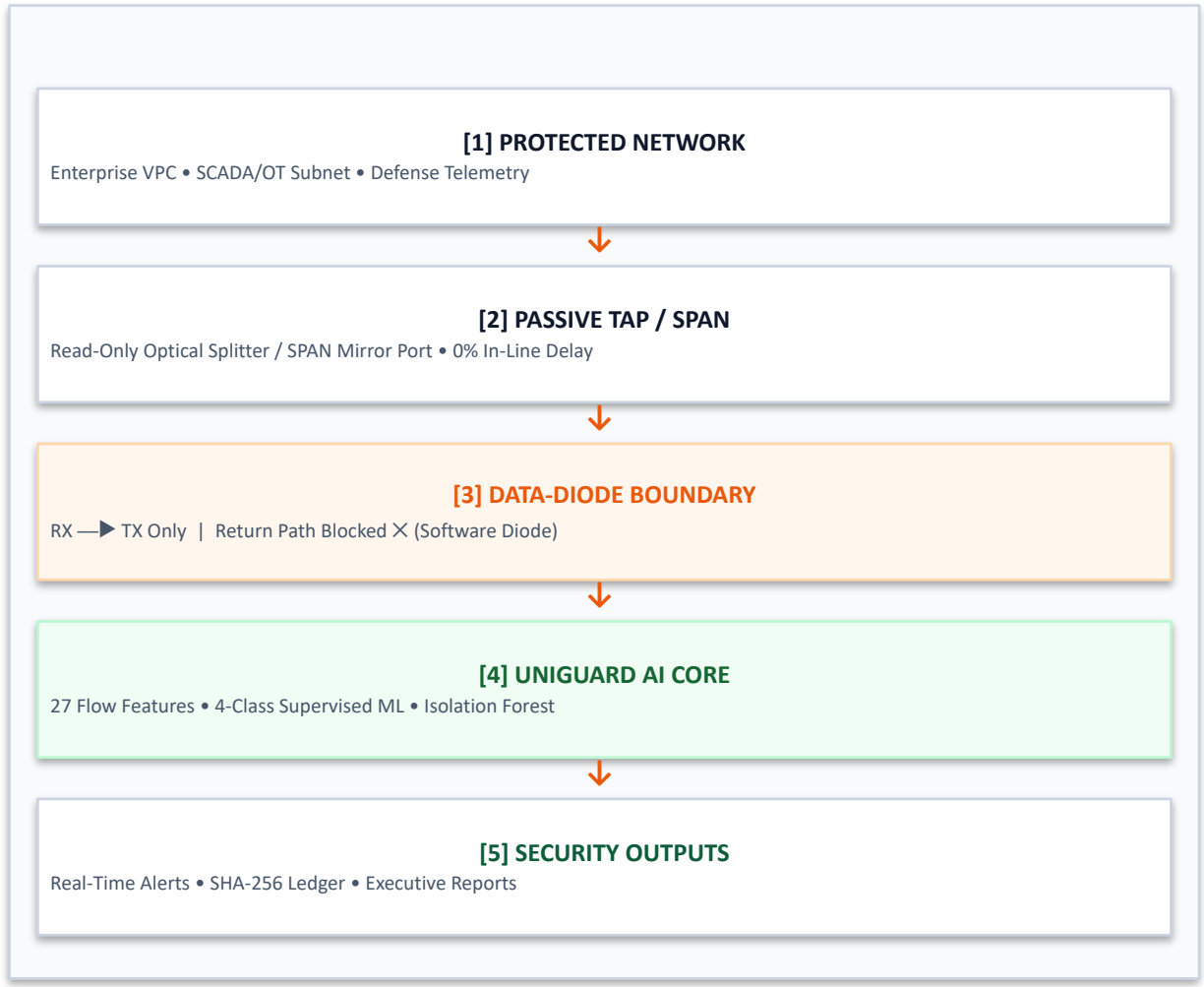
UNIGUARD AI operates strictly on incoming mirrored optical traffic (RX). It enforces 0% outbound packet emission, completely eliminating the possibility of security sensors being hijacked as a reverse attack conduit into critical enclaves.

02 | KNOWN + UNKNOWN — Two-Tier Machine Learning

Features a dual-engine architecture: Tier 1 HistGradientBoosting classifies known volumetric DDoS, port scans, and C2 beacons in <15 ms. Tier 2 Isolation Forest isolates zero-day statistical anomalies without requiring labeled training samples.

03 | VERIFIABLE EVIDENCE — SHA-256 Cryptographic Ledger

Automatically computes chained SHA-256 block digests over 5-tuple flow parameters, Shannon entropy math, and feature matrices, generating tamper-evident forensic records for audit compliance and incident investigation.



TEAM LIQUID

CAL APPROACH

FEATURE STRUCTURED TABLE, TWO-TIER ML FLOW & PRODUCTION TECH STACK

SMART INDIA HACKATHON 2026

END-TO-END PROCESSING PIPELINE:

Authentic PCAP → Magic-Byte Check → Scapy Parsing → 5-Tuple Aggregation → 27 Features → Two-Tier ML → Evidence Ledger → Reports

Feature Category	Count	Deterministic Flow Metrics
Traffic Volume	5	duration, packet_count, byte_count, packets/sec, bytes/sec
Size & Entropy	5	mean_size, std_size, min_size, max_size, Shannon entropy H(X)
Timing / IAT	6	mean_iat, std_iat, min_iat, max_iat, iat_jitter, burstiness
TCP Flags	9	syn, ack, fin, rst, psh, urg, syn_ratio, ack_ratio, rst_ratio

TIER 1: SUPERVISED MULTI-CLASS (HistGradientBoosting)

Classifies: Benign | Volumetric DDoS | Recon Scan | C2 Beaconing

TWO-TIER MACHINE LEARNING CORE

TIER 2: UNSUPERVISED ANOMALY (Isolation Forest)

Novelty Core: Outlier score ≤ 0.0 flags novel zero-day behaviors (Contamination = 0.02)

SPECIALIZED METADATA PROFILERS (DNS DGA & TLS Metadata)

Shannon entropy > 3.4, digit ratio analysis, ClientHello SNI & cipher suite count

PRODUCTION TECHNOLOGY STACK & HARDENING:

- **Frontend:** React 19, TypeScript, Vite 8, Tailwind CSS
- **Backend:** Python 3.11+, FastAPI, Uvicorn, Scapy, DPKT
- **Machine Learning:** scikit-learn, NumPy, Pandas, Joblib
- **Database & Storage:** Supabase (PostgreSQL 15), Row Level Security
- **Security Hardening:** Rate Limiter, Magic Bytes, CORS, SHA-256 Ledger

TEAM LIQUID

LITY AND VIABILITY

ANALYSIS, TARGET USE CASES, CHALLENGE-MITIGATION MAPPING & VERIFIED BENCHMARKS

SMART INDIA HACKATHON 2026

✓ FEASIBILITY ANALYSIS

01 Technical Feasibility: 100% software prototype fully operational with real PCAP binary ingestion.
02 Modularisation: Decoupled FastAPI microservices, streaming ingestion, and swappable ML heads.
03 Market Need: Urgent demand across defense and critical infrastructure for zero-probe security.
04 Economic Viability: Software-first deployment avoids costly proprietary hardware appliance lock-in.

TARGET USE CASES & DEPLOYMENT ENCLAVES

- **Critical Infrastructure:** Power grids, nuclear plants, and water SCADA network monitoring.
- **Defense OT Enclaves:** Air-gapped telemetry networks with strict zero reverse conduit rules.
- **Financial Core Ledgers:** Banking transaction pipelines requiring tamper-evident telemetry.
- **Enterprise Cloud VPC:** Multi-tenant VPC mirror analysis without active probing overhead.

⚠ Identified Operational Challenge	Direct Mitigation Strategy in UNIGUARD AI
Memory Pressure: High memory during massive PCAP file parsing	50 MB upload boundary + streaming chunking avoids memory exhaustion
Strict Passive Boundary: Zero return ACKs complicate TCP tracking	Directional 5-tuple flow aggregation extracts state without ACK handshakes
Zero-Day Threats: Emerging attack signatures absent in initial data	Tier 2 Isolation Forest detects outlier distributions without labeled classes
Encrypted Protocols: Payload opacity in modern TLS 1.3 / QUIC	Non-decrypting ClientHello SNI and Shannon entropy profile encrypted flows

DEPLOYMENT ROADMAP: [CURRENT] Software-simulated diode boundary & microservice prototype (Fully Functional) → [FUTURE / OPTIONAL] Physical optical tap + Hardware data diode (TX LED → RX Photodiode)

59 PASSED	PASS	SHA-256	82.24%	2.8%
0 Failed Tests	Frontend Build	Byte-Verified Checksums	Real AI Accuracy	Benign FPR

TEAM LIQUID

E AND BENEFITS

T PILLARS, TRADITIONAL NIDS VS UNIGUARD AI COMPARISON & BENCHMARK BAND

SMART INDIA HACKATHON 2026

01 | CRITICAL INFRASTRUCTURE DEFENSE (Zero Probing • Safe Operation)

Guarantees zero interference with sensitive power grids, defense telemetry, and nuclear SCADA networks by enforcing strictly passive traffic capture.

02 | NEXT-GEN SOC OPERATIONS (Sub-Millisecond Threat MTTD)

Eliminates manual alert fatigue by automatically classifying volumetric DDoS floods, port sweeps, and C2 beaconing in under 15 ms per flow.

03 | FORENSIC EVIDENCE & AUDIT (Cryptographic Non-Repudiation)

Generates immutable SHA-256 chained hash blocks across flow parameters and entropy distributions, ensuring regulatory audit compliance.

04 | SCALABLE DEFENSE-IN-DEPTH (Software-First Architecture)

Deploys instantly across virtualized enclaves (VPC/containers) with seamless support for high-assurance physical optical data diodes.

Capability Dimension	Traditional NIDS (Snort / Suricata)	UNIGUARD AI Unidirectional Engine
Active Probing Risk	High (Injects TCP resets, active scan beacons)	**Zero (100% Read-Only Passive Listening)**
Reverse Network Path	Bidirectional (Potential reverse attack conduit)	**One-Way Diode (Return transmission blocked ✕)**
Traffic Dependency	Requires stateful 2-way TCP handshakes	**Aggregates single-direction 5-tuple flows**
Threat Detection	Purely static signature matching	**Tier 1 Supervised ML (HistGradientBoosting)**
Zero-Day Anomaly	Fails on unknown attack patterns	**Tier 2 Isolation Forest (Contamination = 0.02)**
Evidence Integrity	Plain syslog / unverified text logs	**SHA-256 Cryptographic Block Ledger**

100% PASSIVE READ-ONLY | 27 DETERMINISTIC FEATURES | 4-CLASS ML | ISOLATION FOREST | 82.24% ACCURACY | 2.8% BENIGN FPR | SHA-256 LEDGER

TEAM LIQUID

CH AND REFERENCES

L FOUNDATIONS, STANDARDS, BENCHMARK DATASETS & RESEARCH METHODOLOGY

SMART INDIA HACKATHON 2026

FOUNDATIONAL RESEARCH & TECHNICAL REFERENCES

01 | NETWORK SECURITY & DATA DIODE ARCHITECTURES

- **NIST SP 800-82 Rev 2:** Guide to Industrial Control Systems (ICS) Security & Data Diodes
- **RFC 793 & RFC 8446:** Transmission Control Protocol & Transport Layer Security (TLS 1.3)
- **IETF Draft:** PCAP / PCAPNG Binary File Format Specifications & Framing
- **Passive Telemetry:** Unidirectional Optical SPAN Mirroring & 5-Tuple Reconstruction

02 | MACHINE LEARNING & ANOMALY DETECTION

- **Ke et al. (2017):** Histogram-Based Gradient Boosting Ensembles for High-Speed Tabular Data
- **Liu, Ting, & Zhou (2008):** Isolation Forest: Fast Tree-Based Unsupervised Anomaly Detection
- **Shannon, C.E. (1948):** Mathematical Theory of Communication (Information & Lexical Entropy)
- **scikit-learn 1.4+:** Production Model Calibration, Cross-Validation, & Robust Scaling

03 | SECURITY, INTEGRITY & BENCHMARK DATASETS

- **FIPS PUB 180-4:** Secure Hash Standard (SHA-256) for Cryptographic Audit Ledgers
- **CSE-CIC-IDS2018:** Authentic LOIC HTTP Flooding & DDoS Network Attack Captures
- **CTU-13 Dataset:** Stratosphere Lab Neris IRC & Botnet C2 Beaconing Benchmarks
- **CIC-IDS2017 & Nmap:** Ares Botnet Command Channels & Port Reconnaissance Slices

04 | IMPLEMENTATION ARCHITECTURE & TOOLING

- **FastAPI Framework:** High-Performance Asynchronous Python Microservice Architecture
- **Scapy 2.5 & DPKT:** Low-Level Binary Packet Decoding & Header Metadata Extraction
- **PostgreSQL 15 / Supabase:** Relational Incident Schema, JSONB Attributions, & Row-Level Security
- **React 19 & TypeScript:** Component-Driven SOC Dashboard with Zero-Error Type Safety

[1] THEORETICAL FOUNDATIONS
NIST SP 800-82 • RFC Specifications • Shannon Entropy Math

↓

[2] SOFTWARE PROTOTYPE
FastAPI Async Engine • Scapy Ingestion • 27-Feature Math

↓

[3] PCAP BENCHMARKING
CSE-CIC-IDS2018 • CTU-13 Neris • CIC-IDS2017 Datasets

↓

[4] MODEL VALIDATION
Two-Tier Architecture: HistGradientBoosting + Isolation Forest

↓

[5] 82.24% REAL AI ACCURACY
2.8% Benign FPR • Zero Cross-Split Capture Leakage

EMPIRICALLY BENCHMARKED MULTI-CAPTURE DATASETS: [1] CSE-CIC-IDS2018 (LOIC HTTP Flooding) • [2] CTU-13 Scenario 1 (Neris IRC Botnet C2) • [3] CIC-IDS2017 (Ares Botnet / Recon)